

Data Analysis

Data Analysis Intro

Department of Economics

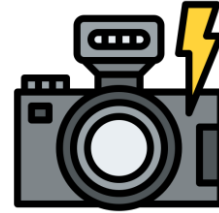
Ivan Bondarev



Email

- i.bondarev@inno.mgimo.ru

Signs



– Please take pictures of the information. Medium priority.

Source: https://www.flaticon.com/ru/free-icon/camera-flash_3460307



– Please write down the information. High priority.

Source: <https://openmoji.org/library/emoji-270D/>

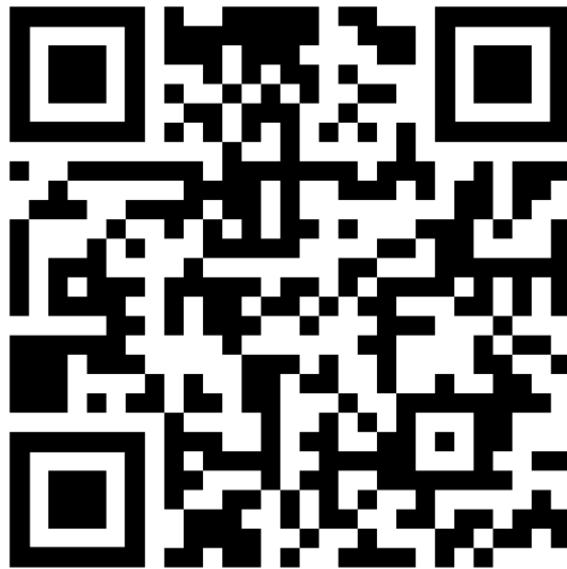
TOPICS

1. Introducing Data Analysis
2. Storing data
3. Jupyter Notebook
4. Python Libraries – pandas
5. Data check

Useful sources

- Github - Nikita Artamonov

<https://github.com/artamonoff>



Useful sources

- Github – Ivan Bondarev

<https://github.com/BondarevIM/MGIMO-SGIA-Data-Analysis>



Data Analysis as a discipline

- *What is Data Analysis useful for?*
- *What basic methods of Data Analysis do you know?*

What's data?

- information, especially facts or numbers, collected to be examined and considered and used to help decision-making, or information in an electronic form that can be stored and used by a computer.

Source:

<https://dictionary.cambridge.org/ru/%D1%81%D0%BB%D0%BE%D0%B2%D0%B0%D1%80%D1%8C/%D0%B0%D0%BD%D0%B3%D0%BB%D0%B8%D0%B9%D1%81%D0%BA%D0%B8%D0%B9/data>

Types of data

- Numerical
- Non-numerical/Categorical:
 - Unordered
 - Ordered
- *What examples of these data types can you provide?*
 - *What are the sources of data?*

Storing data

- .xlsx – an Excel workbook: sheets, formulas, formatting
- .csv – plain text, one line per row
- Both store tables
- Both start with a header row

.csv is the most widespread format for data storage

What is a CSV?

- CSV = comma separated values
- Plain text. One line = one row of the table
- Inside a line, values are split by a **separator**

```
header1,header2,header3
```

```
value11,value12,value13
```

```
value21,value22,value23
```

The separator

- The separator splits values inside a row
- Usually a comma (,) or a semicolon (;)
- Open the file in a text editor and look at line 1
- Wrong separator → the whole row lands in one column

The comma trap

`0.21, 8.68` → two columns

`0,21;8,68` → two columns

`0,21,8,68` → four columns

- Decimal comma → use a semicolon as the separator
- For some countries' formats Excel writes a decimal comma and saves CSV with semicolons

Rules for a clean table

- One column – one type of data; do not write in one column both numerical and non-numerical values
- Leave a cell empty for a missing value
- Do not edit a CSV in Excel without need
 - Excel rewrites dates, drops leading zeros, changes separators
- Keep the original file untouched

Dataset sources

- Our World in Data

<https://ourworldindata.org/>



- Kaggle

<https://www.kaggle.com/datasets>



and many more...

Jupyter Notebook

Jupyter Notebook is an open-source web application used to create and share documents that contain live code, equations, visualizations, and explanatory text.

URL: <https://jupyter.org/>



Jupyter Notebook: .ipynb

- .ipynb is the Jupyter Notebook file format
- It keeps code, results and text in one file
- Cells run one by one, and the order matters
- Put the data file next to the notebook – then your code will target the file directly without a need to specify the full path to the file

Python Pandas

- Pandas is an open-source software library built on top of the Python programming language specifically for data manipulation and analysis.

URL: <https://pandas.pydata.org/>



Loading data with pandas

```
import pandas as pd  
df = pd.read_csv("Diamond.csv")  
df = pd.read_csv("Diamond.csv", sep=";")
```

- read_csv expects a comma by default
- For a semicolon file write sep=";"

What is a dataframe?

- df is short for dataframe
- A dataframe is a pandas table: named columns, numbered rows
- `pd.read_csv()` gives you a dataframe
- The name is a label. Write the name – get the table

Naming a dataframe

- Letters, digits and underscores only
- A name cannot start with a digit
- No spaces and no hyphens
- Names are case-sensitive: df and DF are different
- Do not use sum, list, type, max, str, pd, np

One file – one name

DO NOT WRITE CODE LIKE THIS:

```
df = pd.read_csv("Diamond.csv")  
df = pd.read_csv("Wage.csv")
```

The second line replaces the first table. No warning.

DO WRITE CODE LIKE THIS:

```
df1 = pd.read_csv("Diamond.csv")  
df2 = pd.read_csv("Wage.csv")
```

Better: meaningful names

```
df_diamonds = pd.read_csv("Diamond.csv")  
df_wages = pd.read_csv("Wage.csv")
```

- Two or three words, lowercase, underscores
- Six months later you will not remember what df3 held
- A joke name runs fine, but you will type it fifty times

Reassigning on purpose

```
df = df.dropna()
```

- Normal practice, but the original table is then gone from memory

```
df_clean = df.dropna()
```

- Keeps both the raw table and the cleaned one

Data check: first commands

`df.head()`

- the first five rows

`df.shape`

- number of rows and columns

`df.info()`

- column names and data types

- All data in one column → wrong separator